

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 3 Lesson 1 Exercise 1: More with Relationships

Resolving Many to Many Relationships (S3L1 Objective 2)

After completing the initial ERD it was shown to the client to confirm that the business was being correctly reflected in the design. The following is a transcript from that meeting:

Meeting Transcript

Interviewer: Do you see any major problems with the design of your new system?

Manager: It looks like you have covered everything does that mean that it is nearly finished?

Interviewer: Not quite we still have quite a lot to do before it is a working database system. I have a few more questions for you just to clear some things up before we finalize the design.

Manager: No problem.

Interviewer: A customer can represent a team; does each team have a single named representative or can many people place orders on the team's behalf?

Manager: There can only be one person that is the official representative of the team.

Interviewer: Can the same person represent multiple teams?

Manager: No they can only be assigned to a single team at a time.

Interviewer: At one of our earlier meetings you said that each team representative is assigned a single sales representative. How are the sales representatives assigned to their teams?

Manager: One of the sales representatives is assigned the role of supervisor and they are responsible for assigning individual teams to all the sales representatives including themselves.

Interviewer: You also said that sales representatives sometime handle individual customer complaints?

Manager: Yes if a customer has an issue with their order a sales representative will help them out.

Interviewer: When a customer places an order can that order be transferred to someone else?

Manager: No the order is between the company and the person who placed the order; that can never change.

Interviewer: Can a customer only register a single address as a delivery destination or can they have multiples?

Manager: We allow our customers to register multiple addresses with us so that they can have their orders delivered to the location that suits them best. An address can only be allocated to a single customer.

Interviewer: If you store multiple addresses for the customers do you also store multiple addresses for the sales representatives?

Manager: No we only store the current address information of our staff members, if they move address we update the information there is no need to track their address history.

Interviewer: Going back to customers, can a customer be both a team representative as well as an individual customer?

Manager: Absolutely, but they will have two separate accounts with purchases on one not affecting purchases on the other.

Interviewer: Teams get a discount based on the number of players, how does that work?

Manager: A team needs a minimum of 9 players to receive a discount. If a team has a minimum nine players on their roster they get a 5% discount, if they have a minimum of 25 players they get a 10% discount and for 40 players or more they get a 20% discount. The appropriate discount is allocated to the team.

Interviewer: Is there anything that you are thinking about introducing that might affect the system either now or in the future?

Manager: We are looking at introducing a loyalty card scheme for the individual customers. This would be an optional scheme that would allow loyalty card members to attend special sale evenings where they can purchase items at a reduced price.

Interviewer: Would you want to record what each loyalty card scheme member purchases at these events.

Manager: No the restriction will be that only customers who have loyalty cards will be allowed in the store. We don't need to know what items they purchased.

Interviewer: So you would like us to track the price of items over time so that you can change them?

Manager: Yes that would make life so much easier for us.

In relational databases you can have the following three types of relationship that represent the information requirements and the rules of the business.:

- Many-to-one (M:1) or one-to-many (1:M)
- Many-to-many (M:M)
- One-to-one (1:1)

If you have any many to many (M:M) relationships in your ERD then you will need to resolve those before going any further. Remember you resolve a M:M relationship by:

- Delete the M:M relationship.
- Create a new entity (intersection).
- Create two identifying relationships with the many side on the new intersection entity.
- Identify and create additional attributes in the intersection entity if required. Think about how you would store how many of each item and how many items were actually shipped to the customer.
- Evaluate whether the two existing identifying relationships constitute a UID for the intersection entity or whether an artificial UID has to be created for the new entity.
- Add relationship labels to the new identifying relationships.

Suggested starting point:

